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$$\left\{ \begin{array}{c} 3, \\ 7, \\ 13 \end{array} \right\}$$

$$\left\{ \begin{array}{c} a, \\ p, \\ u \end{array} \right\}$$

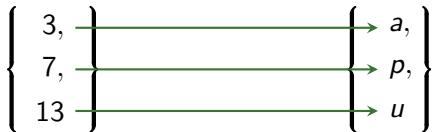
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- Equivalently: there is a bijection between the sets.

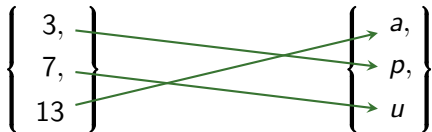
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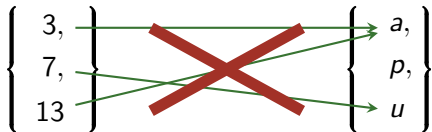
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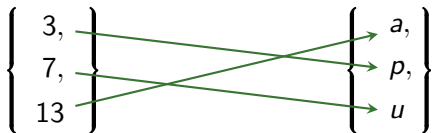
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We use the same approach for infinite sets.

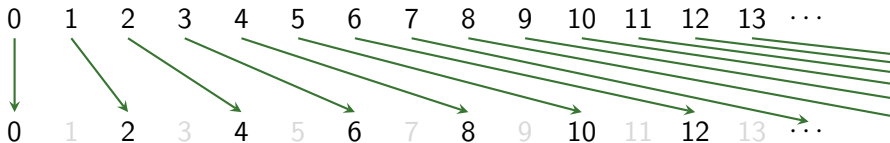
Unintuitive Consequences

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- And the natural numbers and the integers have the same cardinality.
- The natural numbers and Σ^* have the same cardinality.

Unintuitive Consequences

“Two sets have the same cardinality if there is a bijection between them.”

- So, the natural numbers and the even natural numbers have the same cardinality.
- And the natural numbers and the integers have the same cardinality.
- The natural numbers and Σ^* have the same cardinality.
- The latter result also implies that even the natural numbers and the rational numbers have the same cardinality.
- But using techniques shown on the next slide show that the cardinality of the reals is strictly larger than that of the natural numbers.

Back to our Question

- Assume the natural numbers and the subsets of the natural numbers have the same cardinality.
- Then, there is a bijection f between the natural numbers and the subsets of the natural numbers.
- So, the list $f(0), f(1), f(2), \dots$ contains all subsets of the natural numbers.

Back to our Question

- We can arrange the list $f(0), f(1), f(2), \dots$ in an infinite table such that row i represents the subset $f(i)$.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	...
$f(0)$	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	...
$f(1)$	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	...
$f(2)$	✗	✗	✓	✓	✗	✓	✗	✓	✗	✗	✗	✓	✗	✓	...
$f(3)$	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	...
$f(4)$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	...
$f(5)$	✗	✓	✓	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	...
$f(6)$	✓	✗	✗	✓	✗	✗	✓	✗	✗	✓	✗	✗	✓	✗	...
$f(7)$	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Back to our Question

- We can arrange the list $f(0), f(1), f(2), \dots$ in an infinite table such that row i represents the subset $f(i)$.
- Consider the set $\overline{D} = \{3, 5, 7, \dots\}$ obtained by **flipping** the diagonal.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	...
$f(0)$	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	...
$f(1)$	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	...
$f(2)$	✗	✗	✓	✓	✗	✓	✗	✓	✗	✗	✗	✓	✗	✓	...
$f(3)$	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	...
$f(4)$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	...
$f(5)$	✗	✓	✓	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	...
$f(6)$	✓	✗	✗	✓	✗	✗	✓	✗	✗	✓	✗	✗	✓	✗	...
$f(7)$	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Back to our Question

- We can arrange the list $f(0), f(1), f(2), \dots$ in an infinite table such that row i represents the subset $f(i)$.
- Consider the set $\overline{D} = \{3, 5, 7, \dots\}$ obtained by **flipping** the diagonal.
- It differs from every set $f(i)$ in the membership of i .

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	...
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$f(5)$	✗	✓	✓	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	...
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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

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- Consider the set $\overline{D} = \{3, 5, 7, \dots\}$ obtained by **flipping** the diagonal.
- It differs from every set $f(i)$ in the membership of i .
- Thus, $\overline{D}$ is not in the list $f(0), f(1), f(2), \dots$.
- This is a contradiction to our assumption that the list contains all subsets, i.e., our assumption must be wrong: there is no bijection between the natural numbers and the subsets of natural numbers.

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This process, due to Georg Cantor, is called **diagonalization** and has many applications in set theory, logic, and computability.

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The paradox relies on **self-reference**, i.e., “does the barber shave *himself*?”

Can we apply self-reference to Turing machines?

Agenda

1. Warm-up: Hotels and Barbers
- 2. The Halting Problem**
3. Closure Properties

Encoding of Turing Machines

Let $M = (Q, \Sigma, \Gamma, s, t, r, \delta)$ be a DTM. We assume without loss of generality $Q = \{1, 11, \dots, 1^{|Q|}\}$ and $\Sigma = \{0, 1\}$. Also, let $rep_\Gamma: \Gamma \rightarrow \{1, 11, \dots, 1^{|\Gamma|}\}$ be an encoding of Γ such that $rep_\Gamma(0) = 1$, $rep_\Gamma(1) = 11$, and $rep_\Gamma(_) = 111$.

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Then, M is encoded by the word $\ulcorner M \urcorner$ over $\{0, 1\}$ defined as follows:

$$1^{|Q|} 0 1^{|\Gamma|} 0 s 0 t 0 r 0 w_{\delta}$$

where w_{δ} is the list of encodings of transitions. Each $\delta(q, b) = (p, a, d)$ is encoded by

$$q 0 rep_{\Gamma}(b) 0 p 0 rep_{\Gamma}(a) 0 dir(d) 0$$

where $dir(-1) = 1$ and $dir(+1) = 11$.